THE ASSOCIATED GRADED OF A ZETA-STABLE MODULE IS A DIRECT SUM OF SIMPLE MODULES WHEN THE MODULE IS STABLE FOR A GENERIC PARAMETER

HIRAKU NAKAJIMA

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Allowed standard prerequisites (editorial).

- The zeta-bullet-stability of a module (B,a,b), the parameter zeta-bullet and the notion of a generic parameter in the sense of the paper. Defined earlier in the paper (the author's Definition at source lines 746-787 and the subsections on stability, source lines 726-1019). Fixes the hypothesis of the statement and the parameter with respect to which the module is assumed stable.
- The face structure of the set of stability conditions and the decomposition of a polystable module into its stable components. Used inside the counted proof, where the author argues through the faces of the stability parameter and the associated graded object. Supplies the device by which the associated graded of a polystable module is described and why the summands are simple.

Source reference. Original lemma 2.18: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1710–1710. Navigation pointer only: the counted proof appeals to this lemma for the surjectivity of the maps beta_i; the lemma's own statement and proof are not reproduced here and the property it supplies is declared as a named standard prerequisite.

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Source reference. Original theorem 2.11: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1219–1219. Navigation pointer only: the counted proof uses the author's earlier non-emptiness criterion in one step to recognise which simple modules can occur; the criterion itself is not reproduced.

Source reference. Original lemma 2.8: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 999–999. Navigation pointer only: the counted proof refers to the author's face-structure lemma in one step of the face argument; the lemma itself is not reproduced here.

Lemma 0.1. *A $\zeta^\bullet$-stable point $B \in \mu^{-1}(0) \subset \mathbf{M}(\mathbf{v}, 0)$ satisfying the normalization condition $\zeta^\bullet \cdot \mathbf{v} = 0$ is of one of the following three forms*:

(a) *a point in $X_{\zeta^\circ} \setminus \bigcup_{i \in I_0^0} C_i$,*
(b) *$\mathbf{v}$ is a coordinate vector $\mathbf{e}_i$ for an $i \in I_0^0$, or*
(c) *$\mathbf{v}$ is $\delta - \alpha_h$, where α_h is the highest root vector of a connected component of the sub-Dynkin diagram I_0^0.*

In cases (b), (c) *the corresponding $\zeta^\bullet$-stable point is unique up to isomorphisms.*

Proof of Lemma 0.1. Consider the criterion in Theorem 2.11. Since our graph is of affine type, we have $p(x) = 1$ if x is an imaginary root (i.e., $x = m\delta$ with $m \geq 1$) and $p(x) = 0$ if x is a real root. Therefore if $\mathbf{v}$ is imaginary root, the condition is equivalent to $\mathbf{v} = \delta$. This is the case (a). If $\mathbf{v}$ is a real root, then the condition is that $\mathbf{v}$ cannot have a nontrivial decomposition $\mathbf{v} = \sum_t \beta^{(t)}$ with $\zeta^\bullet \cdot \beta^{(t)} = 0$. Then it is clear that we have either (b) or (c).

Next show that a $\zeta^\bullet$-stable point is unique in case (b), (c). It is clear in the case (b). Consider the case (c). We apply an argument used by Mukai for the case of rigid sheaves on a $K3$ surface: Let B, B' be $\zeta^\bullet$-stable points with the underlying I-graded vector space V of dimension vector $\delta - \alpha_h$. We consider the complex

$$\mathrm{L}(V,V) \xrightarrow{\alpha} \mathrm{E}(V,V) \xrightarrow{\beta} \mathrm{L}(V,V),$$
$$\alpha(\xi) = B\xi - \xi B', \qquad \beta(C) = \varepsilon(BC - CB').$$

Since the alternating sum of dimensions is 2, we cannot have both α is injective and β is surjective. If α is not injective, we have a nonzero homomorphism ξ from the module B to B'. Then considering submodules $\operatorname{Ker}\xi \subset B$ and $\operatorname{Im}\xi \subset B'$, the $\zeta^\bullet$-stability of B and B' imply $\operatorname{Ker}\xi = 0$ and $\operatorname{Im}\xi = B'$. Therefore B and B' are isomorphic. If β is not surjective, we consider $\beta^* \in \mathrm{L}(V^*, V^*) \cong \mathrm{L}(V,V)$ instead, we get the same assertion.

Finally consider the case $\mathbf{v} = \delta$. By Lemma 2.8(3b) and the subsequent remark B is ζ°-stable. So B corresponds to a point $[B]$ in X_{ζ°. Therefore it only remains to determine which point in X_{ζ° is $\zeta^\bullet$-stable.

If B is not $\zeta^\bullet$-stable, we consider a Jordan–Hölder filtration to find a proper quotient which is $\zeta^\bullet$-stable. By the above discussion, the quotient must have the dimension vector $\mathbf{e}_i$ with $i \in I_0^0$ or $\delta - \alpha_h$. But in the latter case, B contains a submodule with the dimension vector $\mathbf{e}_i$ with $i \in I_0^0$. This is impossible for the ζ°-stability, as $\zeta^\circ \cdot \mathbf{e}_i > 0$. Thus B has the quotient with the dimension vector $\mathbf{e}_i$ with $i \in I_0^0$. In particular, β above is not surjective, and hence B is contained in the exceptional component C_i by [5, Proof of 5.10]. On the other hand, if B is in C_i, β is not surjective, and it has a quotient with dim $= \mathbf{e}_i$. It is not $\zeta^\bullet$-stable. Therefore $\zeta^\bullet$-stable points are precisely in $X_{\zeta^\circ} \setminus \bigcup_{i \in I_0^0} C_i$. □

References

[1] Crawley-Boevey W., Geometry of the moment map for representations of quivers, *Compositio Math.* **126** (2001), 257–293.

[2] Kac V.G., Infinite-dimensional Lie algebras, 3rd ed., Cambridge University Press, Cambridge, 1990.
[3] Kashiwara M., Saito Y., Geometric construction of crystal bases, *Duke Math. J.* **89** (1997), 9–36, q-alg/9606009.
[4] Nakajima H., Instantons on ALE spaces, quiver varieties, and Kac–Moody algebras, *Duke Math. J.* **76** (1994), 365–416.
[5] Nakajima H., Varieties associated with quivers, in Representation Theory of Algebras and Related Topics (Mexico City, 1994), *CMS Conf. Proc.*, Vol. 19, Amer. Math. Soc., Providence, RI, 1996, 139–157.
[6] Nakajima H., Quiver varieties and Kac–Moody algebras, *Duke Math. J.* **91** (1998), 515–560.
[7] Nakajima H., Lectures on Hilbert schemes of points on surfaces, *University Lecture Series*, Vol. 18, American Mathematical Society, Providence, RI, 1999.
[8] Nakajima H., Quiver varieties and finite-dimensional representations of quantum affine algebras, *J. Amer. Math. Soc.* **14** (2001), 145–238, math.QA/9912158.
[9] Nakajima H., Quiver varieties and tensor products, *Invent. Math.* **146** (2001), 399–449, math.QA/0103008.
[10] Nakajima H., Reflection functors for quiver varieties and Weyl group actions, *Math. Ann.* **327** (2003), 671–721.
[11] Nakajima H., Sheaves on ALE spaces and quiver varieties, *Mosc. Math. J.* **7** (2007), 699–722.
[12] Saito Y., Crystal bases and quiver varieties, *Math. Ann.* **324** (2002), 675–688, math.QA/0111232.